

Multiple Level of Details Neural Implicit Surface Representation for Unconstrained Viewpoint Rendering

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ABSTRACT

To mitigate artifacts and performance degradation in neural implicit representation visualization for unconstrained viewpoint rendering, we propose a feature voxel grid-based neural representation architecture. This approach flexibly encodes the implicit surface with multiple levels of detail, facilitating high-quality rendering with dynamic switching between detail levels. Additionally, we implement a range-limited strategy to concentrate on sampling valid areas while excluding undefined areas. Our results in unconstrained viewpoint scenarios demonstrate the effectiveness of our method. This work extends the capabilities of neural implicit representations, broadening their potential applications beyond previously defined limitations.

Index Terms: Human-centered computing—Visualization—Visualization systems and tools—Visualization toolkits

1 INTRODUCTION

The ability of neural implicit representations to accurately depict detailed 3D geometry has been demonstrated, highlighting their high capacity for visual quality and robustness. NeRF [1] employs a neural radiance field to predict radiance color and density using a multilayer perceptron (MLP). Surface-based neural implicit representations, such as those utilizing neural Signed Distance Function (SDF), offer a viable alternative.

Given 3D coordinates, the SDF measures the distance from a point to the nearest geometry surface, with positive values indicating points outside the surface and negative values representing points within. The surface is then represented by the zero-value set of the SDF. In comparison to volumetric methods, a well-defined surface allows for various operations, such as normal extraction, which is vital for global illumination. However, surface-based methods impose more stringent conditions for network convergence and often require large MLPs, impacting efficiency.

Sampling strategy is another crucial aspect of implicit scene rendering. Neural implicit scene rendering relies on sampling along rays from the camera, accumulating colors to compute pixel RGB values, a process known as ray marching. This method requires dense queries for sufficient 3D resolution to ensure visual fidelity, referred to as volume rendering, which inevitably consumes a considerable amount of time to converge. Hierarchical sampling [1, 5]

is a more prevalent approach, involving iterative batches of importance samplings from coarse to fine after initial uniform sampling. Another challenge arises when sampling points fall into the undefined area, leading to unpredictable network inference results due to insufficient training. This uncertainty complicates the rendering of implicit scenes, especially in unconstrained viewpoint scenarios.

2 ARCHITECTURE AND TRAINING

We adopted NGLOD [3] to design a scalable multi-LOD feature grid-oriented framework. It consists of multi-resolution feature grids, light-weight geometry and color decoders, and a sampling strategy. In the bounding box of the scene, we first allocate same number of cubic grids as preset total LOD number, with exponential increasing resolution. Every grid stores trainable features at the eight vertices of each voxel, where features are shared among neighboring voxels. A set of small MLPs denoted as geometry decoders, be responsible for predicting neural SDF at corresponding LODs. In addition, a color decoder is shared among all the LODs. By encoding latent geometry with the feature, the network size of the decoder is reduced. Our small MLPs contain only one hidden layer.

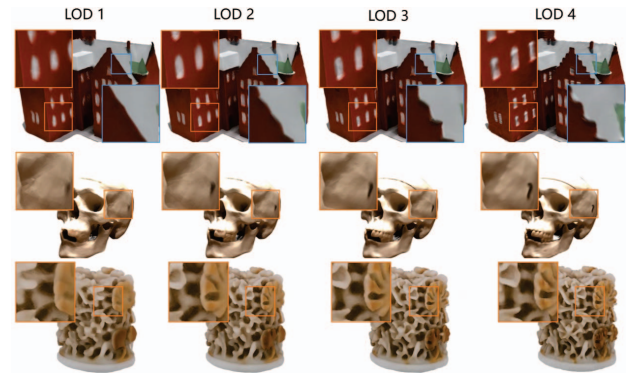


Figure 1: Multiple LOD results on DTU and BlendMVS datasets. From top to bottom are two cases from DTU and one from BlendMVS.

Exhibited as Fig. 1, we equip our architecture with a multi-LOD mechanism to provide dynamic LOD switching. It is evident that as the LOD increases, the fidelity of the geometry improves as well. For low-complexity surfaces, the level of texture detail also benefits from a higher LOD, restoring more high-frequency signals. Given a query point, we first locate all the containing voxels in the grids corresponding to the target and all LODs below it. Then we extract all features by trilinearly interpolating the eight corners of the above voxels. We perform positional encoding (also called Fourier mapping [4]) to the summed feature and input it into the decoders to

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predict the neural SDF and color. Like classic LOD techniques, our architecture can alleviate aliasing during rendering.

Following KiloNeuS [2], we develop a distillation-based point-wise training procedure. Initially, we transfer SDF and color through a pre-trained NeuS model to our architecture. The points outside the grids were discarded directly. However, point-wise distillation is efficient but introducing color biases in result images. As shown by Fig. 2, in the absence of restrictions on accumulated colors, point-wise distillation causes significant disorder appears in the resulting images (top). We introduce an additional ray-wise fine-tune to enforce pixel color constraints (bottom). Subsequently, a pixel-wise fine-tune was conducted on the color decoder to mitigate it while freezing the well-trained geometry decoders. We also test directly reconstructing based on multi-view images to build our model but find it intractable to converge in the foreseeable future. The fine-tune phase leverage the same manner of sampling as regular rendering, with regard to parallel network inference and color accumulation.

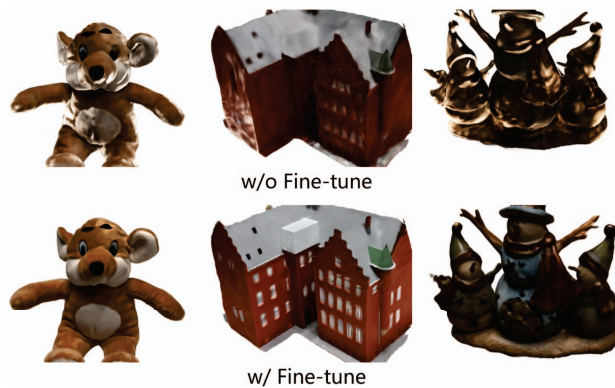


Figure 2: Effect of Fine-tune.

3 SAMPLING STRATEGY

To render a frame, a ray tracer needs to query along rays projected from the camera, accumulating colors and weights to draw pixels. For an unconstrained viewpoint camera however, previous approaches suffer from scale mismatching between implicit geometry and the visible range. We introduce a voxel grid-based strategy that incorporates ray-wise intersection testing before sampling.



Figure 3: NeuS vs. our sampling strategy. From left to right are two cases from DTU and one from BlendMVS.

We delve into the effects of sampling undefined points and acknowledge that directly applying previous sampling techniques to unconstrained viewpoint scenes can result in artifacts and the inefficient use of computational resources. To address this, we tailor an adaptive approach based on the feature grid to allocate sampling points. A ray-AABB test was implemented to retrieve the intersected range of the rays with the model bounding box. Then the renderer takes samplings in the limited ranges on these rays but discards other rays without intersection to save computational resources. Our pipeline then employs iterative hierarchical sampling [1] to obtain query points for the inference phase. Finally, the pixel color is computed through accumulation along the ray; this remains the same as with conventional methods.

Fig. 3 demonstrates a comparison between our work and conventional NeuS in free viewpoint scenarios. NeuS renders using a conventional strategy (top), and our work exploits an adaptive sampling strategy (bottom) from a distance to represent a free viewpoint scenario. It is evident that sampling an undefined area can be unpredictable, especially for models with complex implicit representations. The black pixels in our results denote the discarded rays. The discarding approach significantly boosts the rendering speed when there is a distance between the scene and the camera, a common scenario in free viewpoint rendering. Moreover, for a moving camera, naive NeuS rendering creates dynamic corrugate artifacts, while our strategy produces consistent and higher visual quality results, as exhibited in the demo video: <https://www.youtube.com/watch?v=bW9AMrey6kc>.

4 CONCLUSION AND FUTURE PLANS

We introduce a scalable architecture for neural implicit surface representation, enabling flexible LOD switching and targeting free viewpoint scenarios. We explore the impact of sampling strategies on the rendering results of implicit geometry and present a countermeasure designed to adaptively accommodate a variety of scenarios.

Nevertheless, several limitations persist. The implicit surface is often more challenging to train and achieve higher quality metrics compared to the radiance field. This difficulty translates into reduced robustness for surface methods. In future works, we aim to investigate novel surface representations that offer enhanced stability and superior visual quality while maintaining scalability. We are motivated to explore their capability to depict more intricate structures.

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